

Manufacture of Mark VIII Tanks at Rock Island Arsenal

By

HARRY B. JORDAN

Charter Member A. O. A.

EARLY in the Spring of 1919 there was received at Rock Island Arsenal one of the largest orders in its history. This order covered the assembly of 100 Mark VIII Tanks.

The Mark VIII Tank is a fighting tank, weighing about 40 tons when fully equipped and manned. It carries a complement of 8 men, 1 in the engine room and 7 in the fighting compartment. The 7 men consist of the officer in command, the driver, and 5 gunners; 2 of the gunners manning the 6-pdr. Hotchkiss guns and 3 the Browning machine guns. The tank is equipped with 2 Hotchkiss 6-pdr. guns and 5 Browning machine guns and provides storage for 200 rounds of 6-pdr. ammunition and 20,000 rounds of 30 cal. ammunition. The platforms for the 6-pdr. gun mounts are in sponsons, one of which is located on each side of the tank, pivoting on a vertical hinge partially into the interior of the tank when required, to reduce the overall width.

The motive power is a 12-cylinder Liberty motor developing approximately 400 hp. at 1400 r. p. m. The epicyclic transmission provides for two speeds forward and one speed reverse. In

low gear, with the engine at 1400 r. p. m., the forward speed is 1.4 miles per hour; in high gear, 5.2 miles per hour. The gasoline supply of 150 gallons

allows for ample cruising range. The caterpillar feature in the tank is so designed that no part of the tank extends to the front or to the rear beyond the treads, making it almost nonstallable.

The hull, which is constructed of armor plate, angle iron, steel channels, etc., provides the framework for carrying the engine, transmission, fighting equipment and accessories.

There were no buildings at Rock Island Arsenal specially adapted for the assembly of the Mark VIII tanks on a production basis. The craneway of Shop M (a clear assembly space approximately 60 by 500 feet) equipped with two 15-ton cranes, provided the best space for the hull assembly. In this space it was possible to work on approximately 30 tanks at one time. Each tank occupies a floor space of approximately 410 square feet; i. e., length 34 ft., width 12 ft. While the assembly space in the craneway of Shop M was less than should



Col. Harry B. Jordan
Commanding Officer Rock Island Arsenal

Colonel H. B. JORDAN,
Rock Island Arsenal,
Rock Island, Ill.

June 14, 1920.

Dear Jordan:

I have just received your telegram concerning the completion of the assembly of one hundred Mark VIII tanks on the 10th inst. and I congratulate you on the successful termination of this order.

Sincerely yours,

[Signed]

C. C. WILLIAMS,
Major General, Chief of Ordnance, U. S. A.



Train load of thirty-four cars carrying Mark VIII Tanks

have been available, still the building offered excellent facilities in other respects. It was the only building at the arsenal which had doors wide enough to permit tracks entering one end of the craneway and its cranes offered excellent facilities for moving material from the cars to the floor and for transferring material on the floor. Auto trucks could also unload inside the building.

In addition to handling component parts, the cranes were used for advancing the tanks along the floor during the various stages of assembly. The lack of space made it impossible to complete a tank in place; it necessarily had to be handled on a progressive assembly basis, very similar to that used at most automobile factories. Machine space adjoining the assembly floor was used for the storage of component parts and for a certain part of the sub-assembly work.

The road test followed the assembly, a suitable field located at the southwest corner of the island, approximately three-quarters of a mile from the shops, being utilized. Access to this ground was had by means of two roads, one paved, the other unpaved. An interesting fact developed was that to pay for the repair of the paved road, due to use by tanks, was more economical than to use the unpaved road and pay for the cleaning of the tanks.

Additional Facilities Needed

In order to be able to carry on this work in the craneway of Shop M, it was necessary to move all salvaged machinery and material stored there. The work of moving the machinery was delayed due to the fact that other machinery, declared obsolete, had to be moved before the salvaged machinery could be handled.

Prior to the moving of the salvaged machinery and material, there was only 6,500 square feet available floor space in which to carry on the assembly of the tanks. After all this had been cleared away, approximately 25,000 square feet of floor space was then available.

As the work proceeded it developed that additional floor space must be provided. This space was furnished by taking over parts of the first and second wing and a large portion of the bay of the first floor, adding approximately 15,000 square feet, and giving a total of 40,000 square feet.

The second wing of the first floor was used for the

repair and modification of Liberty motors and transmissions, requiring the installation of overhead trolleys.

In addition to this a number of the sub-assemblies were made in different buildings, as there was no available space for these in the craneway or in the other parts of Shop M. The craneway not being designed for assembly work, it became necessary to install high pressure air lines for the air hammers and wiring for the electric rivet heaters. Then, the electric rivet heaters not furnishing rivets as fast as they were used by the various riveting gangs, it became necessary to install crude-oil furnaces with incidental low-pressure air piping.

Wood horses, trestles and benches were also needed in large quantities for the large force of men engaged in riveting.

Organization

A great deal of difficulty was encountered in building up an organization suitable for handling this class of work, this being the first time work of this nature had ever been attempted at this island.

During the assembly period; *i. e.*, from September 1, 1919, the time when the work really got under way, to June 10, 1920, the date the last tank was completed, the arsenal witnessed a very high percentage of labor turnover on this particular job. During this time there were approximately 3,000 employees engaged upon this assembly work, although at no time were there more than 819 employees actually on the floor; in other words, the labor turnover in a period of approximately nine and one-half months was virtually 400 per cent. It is to be noted that these figures pertain only to the assembly work and not to the personnel engaged upon the manufacture of component parts. The difficulty of making satisfactory progress under such adverse conditions can well be appreciated by anyone familiar with manufacturing work. This tremendous turnover was due to exceptionally high wages paid on the outside (Government adjustments of wages having to follow commercial firms and not precede), and to the fact that knowing that the work at the arsenal would practically cease on June 30, 1920, men were seeking permanent employment elsewhere.

Sub-Assemblies

In planning the work of assembling the Mark VIII tanks, it was advisable to divide the work into various



from Rock Island Arsenal, Ill., to Camp Meade, Md., May 21, 1920

sub-assemblies, allowing the work to be placed on as nearly a production basis as possible; *i. e.*, keeping a certain force of men on each separate, distinct operation.

In the assembly of each Mark VIII tank there are approximately 250 sub-assemblies; and, as previously stated, due to the limited amount of floor space, it was necessary to make some sub-assemblies in various departments throughout the entire Artillery Vehicle Section.

Among the most important of these sub-assemblies were:

- Side doors for main hull,
- Petrol cover doors,
- Ammunition chests,
- Outside tool boxes,
- Peep-hole covers,
- Engine-room doors,
- Switchboards,
- Officers' chests,
- Ball mounts,
- Tin-hat compartments,
- Gauge boards,
- Shell-stop bars,
- Pressure-regulating tanks,
- Combination tap bodies for petrol tanks.

The Liberty motor, when received at this arsenal, was supposed to be ready for installation into the tank, but it developed that, owing to this being an airplane type motor, some modifications would be necessary before the motor would be suitable for the heavy duty to which it was to be subjected in the tank. All were modified by changing from the Zenith to a Ball & Ball carburetor, and by installing in each governor a ball cup and retainer. It was also necessary to remove the distributor heads to synchronize them and to time the cylinders according to the breaker box.

The transmission was not suitable for the tank until some modifications had been made. The insertion of a reinforced internal gear was no short nor simple job.

Order of Assembly

The main hull was the largest assembly and required a force of thirty rivet drivers, thirty buckers and thirty heaters to keep this assembly moving fast enough for the men making the sub-assemblies into the main hull. This work consisted of the assembly of such parts as the girders, hull, side plates, mud chutes, roofs, floors, turrets and road track rollers. The majority of the material mentioned above is $\frac{1}{2}$ -inch

armor plate and angle iron, secured together by approximately 8,000 11-16 inch rivets—a job of riveting requiring great care in many cases. The man and officer's turrets are the second largest sub-assemblies on the tank.

Engine Room Assembly

The placing of the transmission follows the hull assembly. The motor then follows, connecting to the transmission by means of compound clutch (epicyclic type).

The cooling system, consisting of radiator, water tank, sirocco fan and fan drive, is next installed.

The electrical equipment follows, consisting of:

- One 12-volt generator, used for ignition,
- Two 6-volt generators, used for lights, batteries and spark,
- Two 12-volt storage batteries,
- One 6-volt storage battery.

At this time all wiring necessary for the lighting, ignition and control is put in place. These wires are all run through flexible conduit for protection from fire and damage by walking on them or by coming in contact with metal parts about the motor and transmission.

The petrol and air pressure systems are then installed. The petrol system, consisting of three 50-gallon petrol tanks, is located in the rear of the tank, back of the engine room, and is protected from the sides of the tank by means of wood packing strips covered with felt. The flow of gasoline is regulated through a pressure-regulating tank located just over the motor. This practically completes the engine-room sub-assemblies.

Fighting-Compartment Assembly

The remaining sub-assemblies are then confined to the fighting compartment, which consists in the assembling of the gun-hanger bracket for Browning machine guns, ball mounts, periscope and spare barrel clips, peep-hole covers, head rests, periscope holders, semaphores, fire extinguishers, brackets, control units, driver's seat, engine-room doors, tin-hat compartments, officers' chest and various other miscellaneous assemblies.

While the assembly of the sponsons is one of the largest sub-assemblies and one of the first started, the sponsons are not assembled to the tanks until practically the very last. The sponson itself consists of armor plate, angle iron, ammunition storage chests,



Rock Island ruggedness of construction was strikingly demonstrated when this overturned tank, on being righted, started off under its own power

Hotchkiss 6-pounder guns, sponson track rollers and gunners' seat.

Caterpillar Assembly

The driving sprockets are placed on the tank, followed by the track shoes, consisting of two tracks of 78 shoes each. The tracks are assembled to the hull as follows: Four sections comprise the complete assembly; two sections are laid upon the floor, the proper length of the hull. The hull is then raised and set upon these sections. A crane cable is then hooked to one end of another section and raised high enough so that the link can be coupled to the section which is on the floor. It is then drawn along the top of the tank, at the same time lowering the track to the hull. The tracks having been adjusted to proper tension on the hull by the track-adjusting serews, the tank is ready for road test.

Product of Components

Most of the component parts that were to be used in the construction of the Mark VIII tank were purchased by this country from the British Government. These parts consisted of the heavy structural parts of the tank, *i. e.*, armor plate, angle iron, steel girders

and channels, and a large quantity of the equipment carried, such as tools, Hotchkiss guns, camouflage nets, water cans, bird cages, food cans, telescopes, periscopes, festoon lamps, semaphores and various other sundries. All material shipped from England to this country was sent to Raritan Arsenal for temporary storage, for reshipment to Rock Island Arsenal. The balance of the material required in construction was manufactured by various outside contractors throughout the United States and was also shipped to Raritan Arsenal. These items were the Liberty motors, transmissions, compound clutches, petrol tanks, radiators, electrical equipment and front control units.

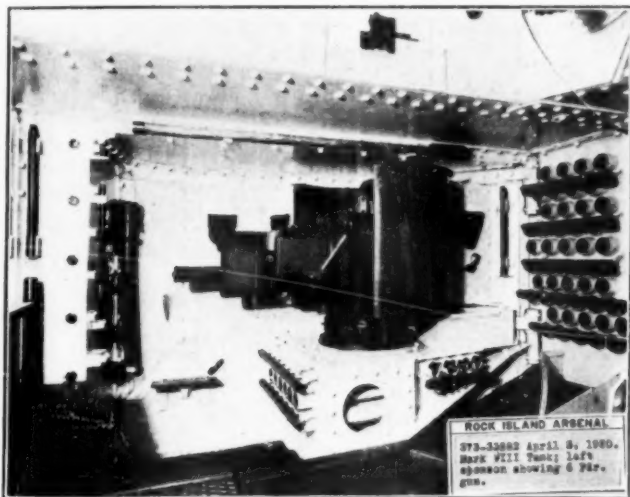
Upon the first inventory of material received it was found that only 65 per cent of the necessary material was on hand, and Raritan Arsenal reported that many of the essential parts needed had not been received at that arsenal from the contractors. It then became necessary to send men to a number of these contractors throughout this country to urge completion of the work, to trace and have the shipment of these parts expedited. Due to congested freight conditions, strikes and other causes it was impossible to

procure some of these parts in time to be of any assistance in the assembly and, therefore, a number of parts were manufactured here to prevent delay in production.

It was necessary to purchase bolts, rivets, lock washers, pipe fittings, angle iron, gauges, tubing, raw material and numerous other items in large quantities. The rivets used in most of the assemblies were an odd size, not in common use in this country, and it was a difficult matter to secure proper stock for manufacture.

Manufacture of Components

As stated above, approximately 35 per cent of the parts entering into the construction of the Mark VIII



tanks had to be manufactured at this arsenal. These consisted of such parts as outside tool boxes, unditching angles, roller sprockets, roof, side, front and rear towing brackets; sponson hinges on hull, engine-control rods, battery shelf assembly, shell tubes, combination tap bodies, petrol-tank packing, track adjusting screws, bolts, nuts, cap screws and pins.

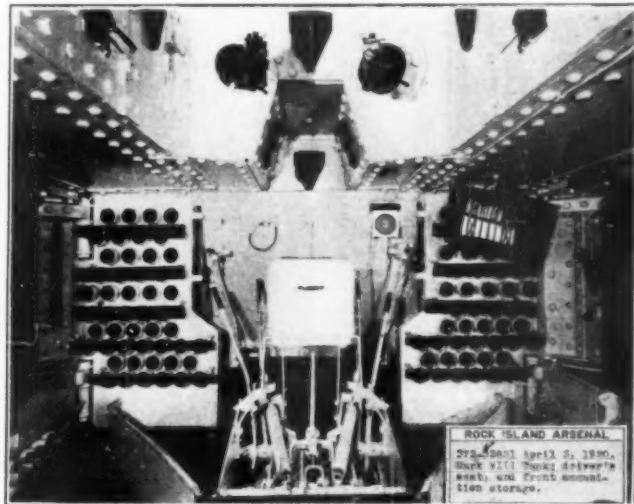
In the manufacture of these items it was necessary to make special tools, jigs and fixtures in order to get production. It might be well to state, however, that most of the parts which were manufactured at this arsenal for the first 25 tanks were made without the assistance of any special tools, jigs or fixtures of any kind, in order to avoid delay in production; and in this way it gave a better idea as to just what tool, jig, or fixture would be best adapted for the work. This also gave time to have these tools, jigs and fixtures manufactured so that they could be used in the balance of the material required.

The items of bolts, nuts, cap screws and pins, as listed above, was probably one of the largest items which had to be contended with, as this quantity ran into the hundreds of thousands. The manufacture of bolts for this job alone kept approximately 25 mechanics and 30 turret lathes and automatic screw

machines busy for a period of about four months. To better explain the quantity required, there was one item of $\frac{5}{8}$ -inch tap bolts, Whitworth thread, of which there were 20,000 required for the manufacture of these Mark VIII tanks.

A quantity of the armor plate was received without heat treatment, and it was necessary that this armor plate be heat-treated before it could be used in the assembly of tanks. As this was practically the first time any work of this nature had been attempted at this arsenal, it was necessary to do a certain amount of experimenting to produce satisfactory results.

Practically all of the armor plate, steel angle iron, steel girders, etc., received at this arsenal had to be straightened to conform to drawings. This required



a force of approximately 50 sheet-metal straighteners and helpers. These men were employed continually on this work during the entire period of construction.

As much of this material was manufactured from castings, it was necessary to keep a force of pattern-makers and molders continuously on this work.

There was a large amount of wood work necessary for these tanks, such as packing for petrol tanks, switchboards, gauge boards, shell-tube boards, map boards, Browning machine-gun rest blocks and officers' chests.

Tests

On February 2, 1919, a sample Mark VIII tank was received at Rock Island Arsenal from Bridgeport, Conn. After remaining at this arsenal for a few days it was ordered out for exhibition purposes to Clinton, Iowa; Iowa City, Iowa; Ottawa Beach, Mich., and Denver, Colo. It was then returned and sent to Savanna Proving Ground, Ill., where it received a road test of 2,155 miles.

A force of men familiar with the handling and operating of the Mark VIII tank were transferred from the Rock Island Arsenal to Savanna Proving Ground to handle this road test, which was conducted to determine any defects in the design of the working parts

of this tank. The most serious trouble developed by this test was the stripping of the teeth in the sprocket roller pinions. It also developed that the exhaust system designed and in use on this tank was not suitable; that the Zenith carburetor originally designed for the 12-cylinder Liberty motor of the aeroplane design was not suitable for this class of work and would have to be replaced with a Ball & Ball carburetor; that the V-belt driving the sirocco was unsatisfactory, calling for a replacement with a 3-inch leather belt, necessitating the changing of the V-pulleys to cast-iron flanged pulleys.

The design and material used in the petrol tank caused considerable trouble due to rust and scale getting into the carburetor. This caused some back-firing through the carburetor; and in a number of instances fires occurred, damaging the electrical equipment.

This road test was given only to the original tank shipped from Bridgeport; but each tank was given a four-hours' shop test to determine the satisfactory operation of the motor and to make any necessary adjustments. Upon the completion of this shop test, the tank was then placed on a road test for ten hours; after the road test it was carefully gone over and inspected, cleaned, painted and prepared for shipment.

Difficulties Encountered

In addition to difficulties mentioned under the heading of tests, the following are a few of the more serious difficulties encountered:

(a) On practically all the material which had to be riveted together it was necessary to drift the holes, due to inaccurate drilling, or to the expansion or contraction of these plates during the process of heat-treating. It became necessary in many cases to weld these holes and to redrill.

(b) In many instances cracks appeared in the armor after the plates were riveted in place. It was then necessary to remove these plates, necessitating the cutting out of all rivets. These cracks were so slight that they could not be detected before assembly and only appeared due to impact of the riveting hammers.

(c) When attempt was made to assemble the Hotchkiss guns to the sponsons, the base was found to be out of true, and it was necessary to remove them from the mounts and straighten them in hydraulic presses.

(d) Due to faulty design and construction of the petrol tanks, leaks were often discovered after assembly, and much trouble and delay was encountered in attempting to keep up the necessary pressure required to send gasoline to the carburetor.

(e) The revolving peep-hole plates, made of armor plate and about 6 inches in diameter, when received

were so warped that they had to be surface ground on both sides before they could be used. There were 22 of these required for each tank, making a total of 2,200.

Incidents of Construction

The construction of the first Mark VIII tank was started July 1, 1919, and the last tank was completed and ready for road test on June 5, 1920, making a total of 286 days to complete the 100 tanks. The first tank was completed on January 5, 1920, and the last just five months later, making a total of 150 days used in the assembly of the tanks after getting on a production basis, or on the average of one tank every 36 hours on the basis of working 24 hours per day. The general assembly on the tanks did not start until about September 1, 1919, due to the fact that a great deal of time was spent in checking in material and waiting for the receipt of the various component parts which were to be shipped to this arsenal by Raritan Arsenal.

The greatest number of tanks which were produced in one week of six days was seven. The greatest number of tanks under construction at any one time was twenty-eight.

On November 13, 1919, the Tank Section was divided into three 8-hour shifts; 50 per cent of the men being employed on day shift, 30 per cent on the first night shift and 20 per cent on the second night shift.

The only "near" serious accident during the construction of the tanks occurred on March 22, 1920, while loading one of the tanks for shipment. The sill of the flat car gave way, allowing the tank to turn turtle off the car. The only damage resulting from this accident was a slight dent in the roof of the main turret and the bending of the semaphore, which damages were very easily repaired without removing the parts from the tank. The driver and mechanic, who were in the tank at the time, were in no way injured other than being slightly shaken up. The tank was righted by coupling another tank to it with towing cables and rolling it completely back on its track, after which it proceeded to the shop on its own power for necessary repairs.

On March 29, 1920, during road test of a tank at the testing ground, near the west end of the island, one of the tanks was run too close to the water's edge and in some manner slipped off, burying itself in the mud and water. The river stage was unusually high at this time of the year, and water coming in contact with the ignition system made it impossible to get the tank out with its own power. It was towed from the river by coupling two Mark VIII tanks together.

While under road test a number of tanks caught fire, through back-firing through the carburetor. These fires caused very slight damages and were easily

extinguished by the use of fire extinguishers supplied to the tanks for any emergency of this nature. There was no one at any time injured or seriously burned in any fires of this nature.

The greatest number of men employed at any one time in the Tank Section was 819. Of this number there was approximately 10 per cent absent each working day. Due to the large labor turnover of employees in this section, a great deal of time was expended in instructing these men in the various operations entering into the construction of the Mark VIII tanks.

The following are averages of the different classes of labor used in the construction:

15 foremen and assistant foremen,

25 sheet-metal straighteners,	
45 machinists,	30 heaters,
65 specialists,	100 assemblers,
30 riveters,	300 skilled laborers,
30 buckers,	200 laborers,

8 clerks in the office personnel, which included stenographer, follow-up men and messengers.

The successful completion of the job on schedule time was due to the exceptional interest in the work and the loyalty and co-operation of the foremen and permanent workers on the job.

The completion of these tanks was promised the Chief of Ordnance by June 15, 1920. It gave the writer great pleasure to inform General Williams of the completion of the last tank on June 10, 1920.

News of the Districts

(Continued from page 26)

M. Huxley, of Wilkinson & Huxley, Chicago, Ill., acting as Secretary.

Springfield Armory District

The territory formerly comprising the Bridgeport Ordnance District now comprises the Springfield Armory District of the Association.

District Headquarters are at the Springfield Armory, and every effort is being made to keep the District Membership interested in Armory affairs.

A Membership Campaign under the leadership of Col. Charles H. Tenney, President of the Springfield Gas Co., has recently been inaugurated and is yielding a highly satisfactory increase in the District Membership.

The other members of the Committee are Henry Brewer, Vice-President, Winchester R. A. Co.; Clifford D. Cheney; W. H. Eaton, Treasurer, Eaton, Crane & Pike; John P. Elton, Treasurer, American Brass Co.; A. W. Erdman, Pratt & Whitney; B. A. Franklin, Director, Strathmore Paper Co.; E. O. Goss, General Manager, Seovill Manufacturing Co.; Charles R. Hare, Production Manager, New Britain Machine Co.; Elmer E. Neal, Works Manager, Marlin-Rockwell Co.; Andrew L. Riker, Vice-President, Locomobile Co., and Samuel M. Stone, Vice-President, Colts' Patent Fire Arms Manufacturing Co., with Col. Julian S. Hatcher acting as District Secretary.

Boston District Activities

The Boston District Membership Committee of the Association, headed by Levi H. Greenwood, of Heywood Brothers & Wakefield Co., as Chairman, have increased the District Membership over 200 per cent.

The other members of the Committee are John W. Higgins, Worcester Pressed Steel Co.; Thomas F. Brown, Industrial Engineering Corporation; Thomas B. Doe, U. S. Cartridge Co.; Stuart W. Webb, Bond & Goodwin; John S. Holbrook, Gorham Manufacturing Co.; Harry C. Dodge, Gray & Davis, Inc.; Alvan T. Simonds, Simonds Manufacturing Co.; James C. Heckman, United Drug Co., with Fred A. Rogers, formerly Major, Ordnance Department, as Secretary.

Notes

Railway Artillery.—The outstanding features in the development of railway artillery in the United States during the last two years lies in the construction and test of the 14-inch gun railway mount, model "E." This gun and mount were designed in the Ordnance Office; the gun was built at Watervliet Arsenal, N. Y., and the mount at Watertown Arsenal, Mass. Gun and carriage tests at the Army Proving Grounds, and service tests at one of the seacoast fortifications have proven this mount to be one of the most serviceable pieces of railway artillery that we have.

The mount may be fired as a sliding mount from any well-ballasted track, but without any specially prepared foundations; it is provided with several degrees of traverse, so that a curved track is not necessary, as in the case of the usual type of sliding mount where no traverse is provided. The carriage is equipped with a base plate suitable for setting on bare ground or on a concrete plug, from either of which the gun can be fired in any azimuth. In target practice, at both long and short ranges, where moving targets giving the extreme rate of change in direction were employed, the practical use of this mount was well established, the general operation, rate of fire and percentage of hits being most gratifying.

Infantry Howitzer.—The Infantry accompanying guns, which were tested at Aberdeen Proving Ground, were manufactured by the Ordnance Department, the two types being copies of Italian designs examined by the Caliber Board in Europe. The first design completed was a 70-millimeter gun, firing a 6-pound projectile; the second design was a 75-millimeter gun, firing an 8.5-pound projectile. The object of manufacturing these two types was to provide models at an early date, in order to crystallize ideas and opinions on what characteristics this type of weapon should embody. The tests have clearly shown that these two models are not satisfactory to fulfill the specifications which have been proposed; *i. e.*, that this weapon shall be suitable as a substitute for both the 37-millimeter gun and the 3-inch trench mortar.

The Ordnance Department is now preparing designs of a weapon that will fire a 6-pound shell for direct fire, and a 10-pound shell for indirect plunging fire. The tests of the two Italian types have been suspended pending a metallurgical investigation which was found necessary.